BERNSTEIN
SOCIETE GENERALE GROUP

## China Pharma and Biotech

# The Long View: Comparing R&D muscles of China Pharma & Biotech


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It's been a while since our last deep-dive in R&D capabilities - we provide an update here with three sets of analysis to assess R&D efficiency across companies, and **Innovent, Akeso, and Hengrui** emerge as future champions.

**Economic return on R&D.** This is the most important measure as it looks at the ability to turn R&D spending into revenue. To evaluate the return on company level, we need to compare revenue from innovative drugs to the cumulative R&D spending a few years back. At the time of our previous assessment, most companies had been too "young" without enough years of R&D data. Now we can get to the return from two angles: 1) sales return on R&D and license income return on R&D (see detailed formula in Exhibit 1). Results: 1) **Leading biotech companies have seen a moderate increase in their sales return on R&D from 2023 to 2025, while pharma players, except Hansoh, suffered a decline at the same time as their top products declined.** 2) In 2025, Hansoh had the highest sales return on R&D at 2.0X across companies. Kelun-Biotech outperformed in license income return on R&D at 1.0X. 3) **Going forward, we expect Akeso and Innovent to improve the most, becoming champions in both sales (~3X) and license income returns (0.7-0.9X) in 2030E. Hengrui ranks best among pharmas in 2030E.**

**R&D efficiency - timeline and conversion. 1) Time-to-market: we calculate the median time between IND application to marketing approval for all innovative products.** The average is 2100 days for oncology products and 2900 days for non-oncology for our coverage. In both categories, Zai Lab has the shortest time-to-market, not surprising given their business model of in-licensing late-stage global products that only require Ph3 or bridging studies in China. Among companies with substantial internal pipelines, **Innovent consistently leads in time-to-market, 1067 days in oncology (~50% of average) and 2111 days in non-oncology (~30% lower than average).** 2) **Hit rate: as evaluated by the total number of marketed innovative drugs over all INDs submitted before end of 2023**, measuring the conversion along the clinical trials funnel. Biotech have almost twice the hit rate as pharma on average, and **Innovent and Akeso again stand out with the highest hit rates at over 40%.**

**Size and quantity measures.** 1) Pipeline size: Hengrui still dominates with quantity with 360+ assets. Notably Innovent now has over 100 assets, entering the level of mature biopharma; also highest potentially FIC % at 28% vs. average c. 20%. 2) No. of trials: Hengrui and Innovent have sponsored the most trials so far. Interestingly, we looked at no. trials sponsored per billion RMB of R&D spending, and Akeso and Hengrui are the most efficient, averaging ~20 trials per year.

## BERNSTEIN TICKER TABLE

| Ticker | Rating | Cur | 20 May 2026 Closing Price | Price Target | TTM Rel. Perf. | Cur | Reported EPS 2025A | Reported EPS 2026E | Reported EPS 2027E | Reported P/E (x) 2025A | Reported P/E (x) 2026E | Reported P/E (x) 2027E |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 9926.HK (Akeso) | M | HKD | 115.00 | 130.00 | 1.4% | CNY | (0.60) | (0.52) | 0.70 | (165.5) | (193.8) | 143.2 |
| ONC (BeOne) | O | USD | 297.01 | 412.00 | 2.1% | USD | 2.63 | 5.65 | 8.93 | 112.9 | 52.5 | 33.3 |
| 1093.HK (CSPC) | M | HKD | 7.39 | 10.70 | (20.8)% | HKD | 0.37 | 0.52 | 0.56 | 17.4 | 12.4 | 11.5 |
| 3692.HK (Hansoh) | O | HKD | 33.60 | 47.00 | (2.9)% | CNY | 0.93 | 1.01 | 1.10 | 31.4 | 28.9 | 26.5 |
| 1801.HK (Innovent) | O | HKD | 79.40 | 120.00 | 5.4% | HKD | 0.48 | 0.71 | 2.90 | 163.7 | 111.9 | 27.4 |
| 600276.CH (Hengrui) | O | CNY | 50.95 | 71.00 | (44.6)% | CNY | 1.19 | 1.54 | 1.92 | 42.8 | 33.0 | 26.6 |
| 6990.HK (Kelun-Biotech) | O | HKD | 437.00 | 526.00 | (8.1)% | CNY | (1.66) | (2.75) | 0.85 | 36.9 | 37.5 | 23.1 |
| 1177.HK (Sino BioPh) | M | HKD | 5.17 | 7.90 | (14.5)% | CNY | 0.19 | 0.20 | 0.22 | 23.5 | 22.7 | 20.5 |
| 9688.HK (Zai Lab) | M | HKD | 14.35 | 15.00 | (81.5)% | USD | (0.16) | (0.15) | (0.09) | 3.6 | 3.3 | 2.8 |
| ASIAX | | | 1,894.75 | | | | | | | | | |
| SPX | | | 7,353.61 | | | | | | | | | |

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

6990.HK, 9688.HK valuation is EV/Sales (x); 9926.HK, 1093.HK, 1177.HK base year is 2024;

Source: Bloomberg, Bernstein estimates and analysis.

# DETAILS

## FRAMEWORK FOR RETURN AND EFFICIENCY METRICS

EXHIBIT 1: **Framework and methodology to evaluate R&D capabilities across companies**

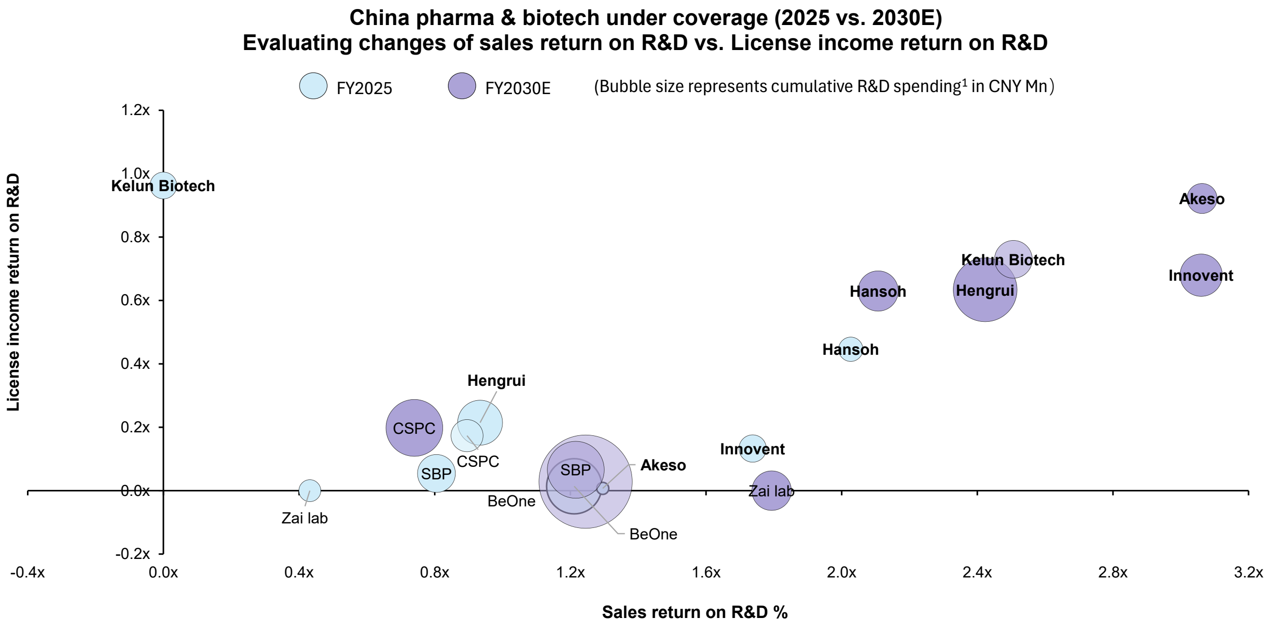
| I. Economic return | II. Efficiency | III. Size & quantity |
|---|---|---|
| **Sales return on R&D**<br>Innovative drug sales in year T / cumulative R&D expense in years T-7 to T-4 (4 years cumulative) | **Time to market**<br>Median time between IND application to marketing approval | **Pipeline size**<br>No. assets at each preclinical and clinical stage |
| **License income return on R&D**<br>License income in year T / Cumulative R&D expense in years T-5 to T-3 (3 years cumulative) | **Hit rate**<br>No. marketed innovative drugs / No. INDs until end of 2023 | **No. trials**<br>No. of trials per year<br>&<br>No. trials sponsored per Bn Yuan of R&D expense |

Source: Bernstein analysis

## CONCLUSIONS FIRST: RETURN ON R&D SPENDING

We estimate a clear shift in China pharma/biotechs R&D return dynamics by 2030E (Exhibit 2). In 2025, Hansoh delivers the highest sales return on R&D, followed by Innovent, while Kelun Biotech leads peers in license income return despite minimal innovative drug sales. Looking to 2030E, Akeso and Innovent are expected to outperform peers across both innovative sales and licensing income-derived R&D returns, driven by the rapid sales ramp-up for their innovative assets and rising milestone/royalty payouts from out-licensed assets. Hengrui emerges as the top-performing pharma in overall R&D returns compared to other pharma companies, underpinned by strong commercial ability and steady licensing momentum.

EXHIBIT 2: **We expect top-tier innovators (Hansoh, Hengrui, Akeso, Kelun, Innovent) to significantly improve sales and licensing returns on R&D by 2030**

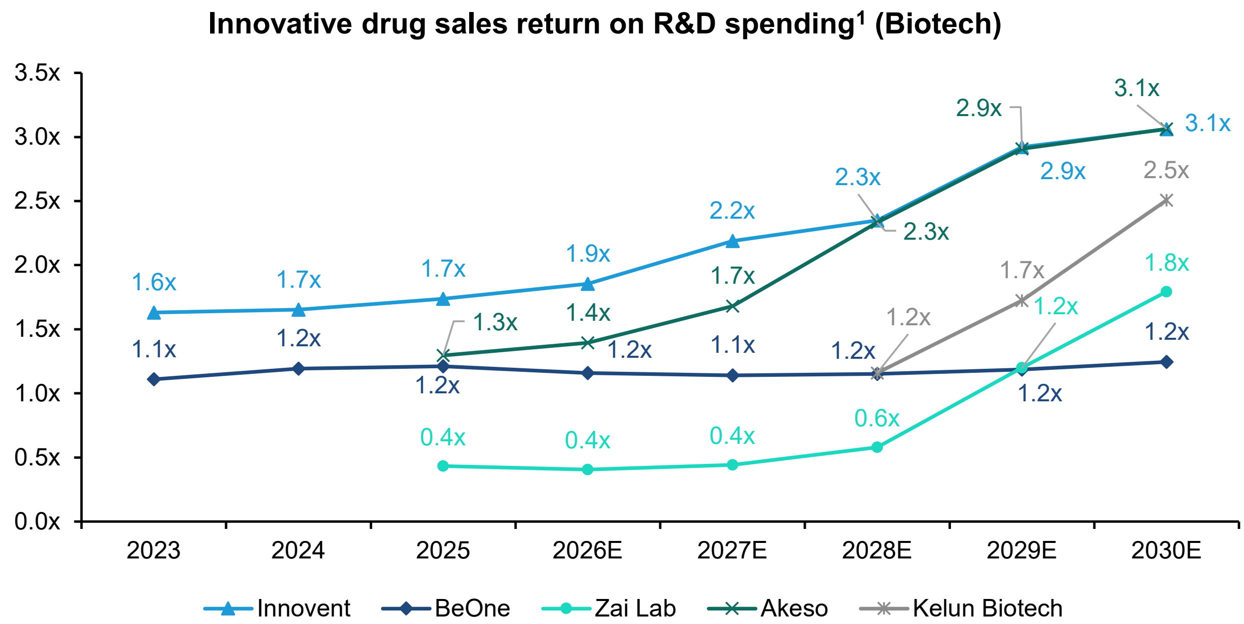



Note: 1. FY2025 cumulative R&D spending is the sum of 2016-2020, while FY2030E is the sum of 2021-2025, except for Kelun Biotech (sum of 2021-2025 and 2026-2030)
Source: Company disclosures, Bernstein analysis and estimates

## RETURN ON R&D: INNOVATIVE DRUG SALES OVER CUMULATIVE R&D EXPENSES

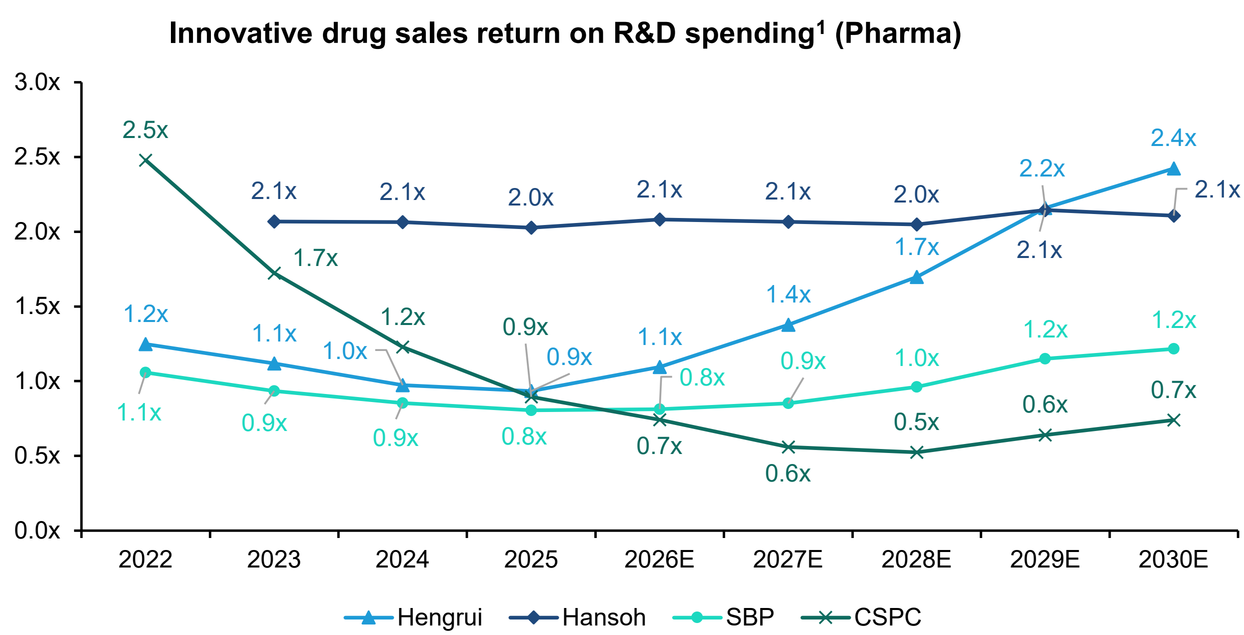
Our analysis measures innovative drug sales return on R&D as **innovative product sales divided by 4-year cumulative R&D spending incurred4 years prior to the sales reporting year**, capturing lagged R&D payback. Across covered biotechs, we estimate **steady, accelerating innovative sales returns on R&D for Innovent and Akeso by 2030E**, supported by their robust domestic commercial ramp-ups alongside controlled R&D spending growth; by contrast, BeOne is expected to experience flat returns as revenue growth modest post Brukinsa uptake while R&D outlays continue rising (Exhibit 3). For covered pharma companies, **Hansoh maintains the highest near-term R&D sales return despite with limited upside ahead**, while broader pharma companies faced declining returns from 2022-2025 as major products enter the later stage of life cycle and ballooning R&D spending. In the long-term, **Hengrui stands out as a late-cycle winner**, with accelerated new product launches driving higher R&D returns from 2027E-2030E (Exhibit 4).

EXHIBIT 3: **We project accelerating innovative sales return on R&D across covered biotech driven by strong commercial ramp-up, contrasting with more mature, stable returns for BeOne**




Note: 1. R&D spending represent 4-y cumulative figures ending 4-y prior to the year of product sales figures
Source: Company disclosures, Bernstein estimates and analysis

EXHIBIT 4: **We forecast a steady recovery for Hengrui to 1.7x by 2028E backed by its innovative drug commitment; in contrast, we foresee CSPC facing headwinds as it navigates a transition phase marked by high R&D ahead of meaningful innovative sales**

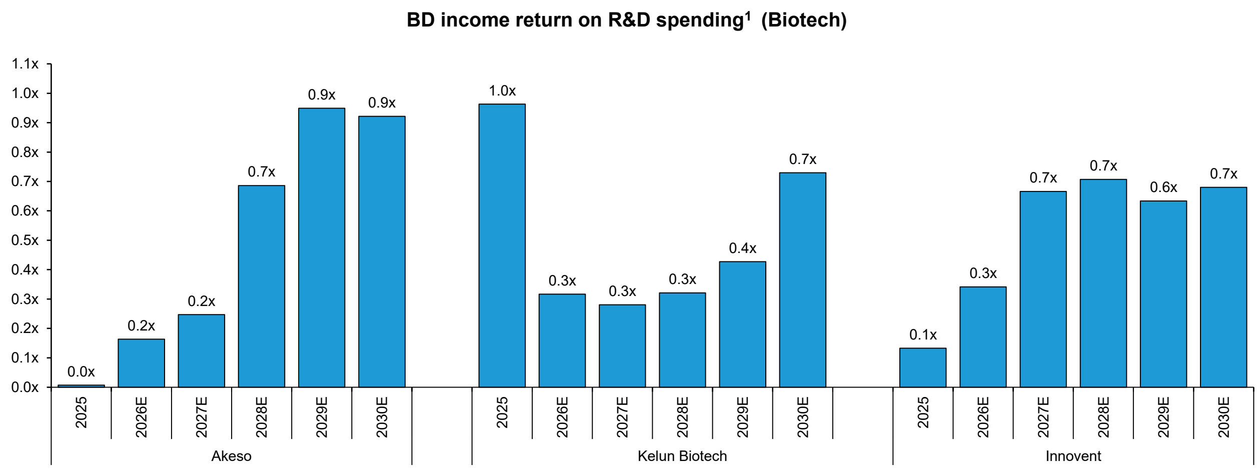



Note: 1. R&D spending represent 4-y cumulative figures ending 4-y prior to the year of product sales figures
Source: Company disclosures, Bernstein estimates and analysis

## RETURN ON R&D: LICENSE INCOME OVER CUMULATIVE R&D EXPENSES

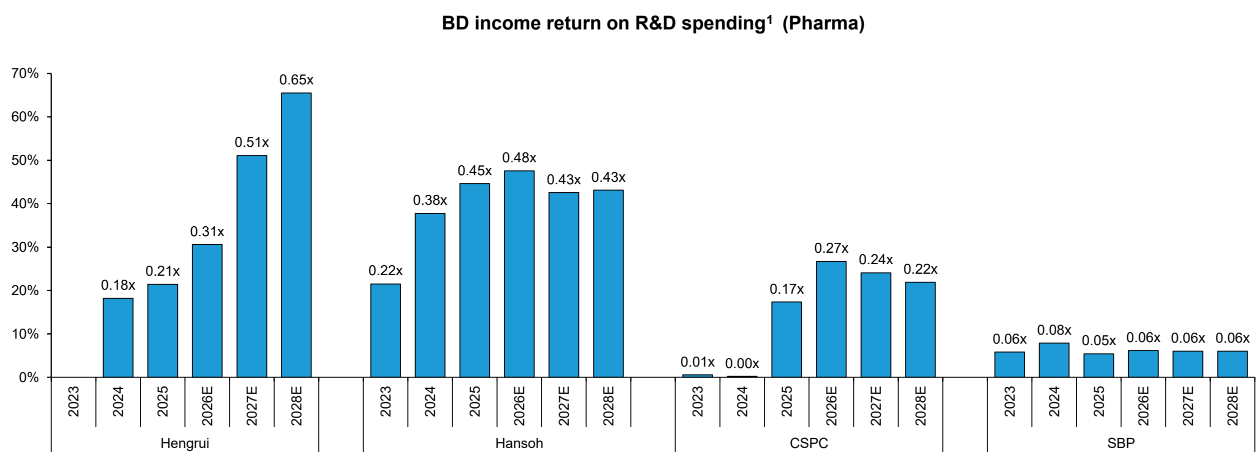
Our methodology calculates BD income return on R&D as **licensing revenue divided by 3-year cumulative R&D spending incurred 3 years prior**, tracking lagged payoff from global partnership investments. For covered biotechs, we foresee out-licensing to remain a core monetization driver. Specifically, **Akeso is expected to lead** as its AK112 will enter a milestone and royalty harvest phase soon following potential FDA approval in 2027E. Kelun Biotech is anticipated to face a near-term return drop as its core asset sac-TMT still in global Ph3 trial development. In addition, we expect **Innovent to see a sharp licensing return jump in 2027E** from potential milestone received from Takeda (Exhibit 5). For covered pharma companies, we anticipate **Hengrui to deliver steadily rising BD income return on R&D given its consistent BD activities and a deep pipeline portfolio**. By contrast, CSPC and SBP are expected to have materially lower returns due to limited BD activities (Exhibit 6).

EXHIBIT 5: **We expect out-licensing to remain a vital monetization engine for covered biotech, with Innovent and Akeso leading the cohort**




Note: 1. R&D spending represent 3-y cumulative figures ending 3-y prior to the year of product sales figures; we omit BeOne Medicines and Zai Lab since their business models are not out-licensing focused
Source: Company disclosures, Bernstein estimates and analysis

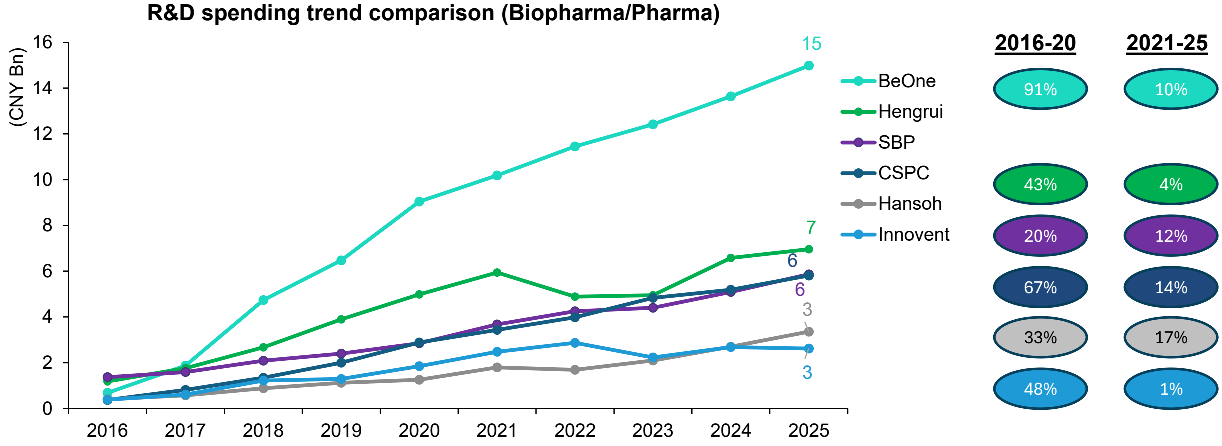
EXHIBIT 6: **Among covered pharma, we estimate Hengrui and Hansoh to lead the cohort in BD return on R&D driven by their proactive out-licensing business model**




Note: 1. R&D spending represent 3-y cumulative figures ending 3-y prior to the year of product sales figures
Source: Company disclosures, Bernstein estimates and analysis

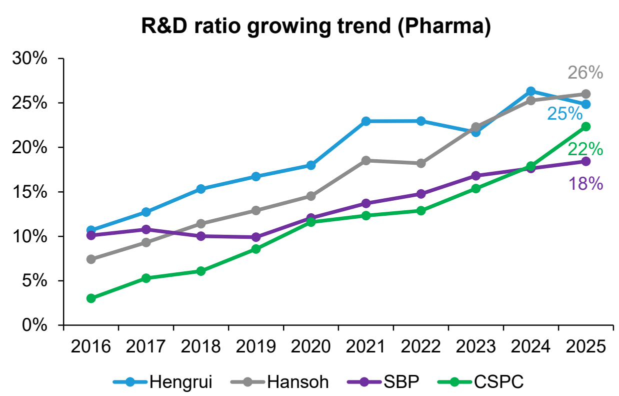
## R&D SPENDING OVER TIME

EXHIBIT 7: **Biopharma (Innovent, BeOne) sees accelerated R&D growth 2016-20 with slowdown in 2021-25**



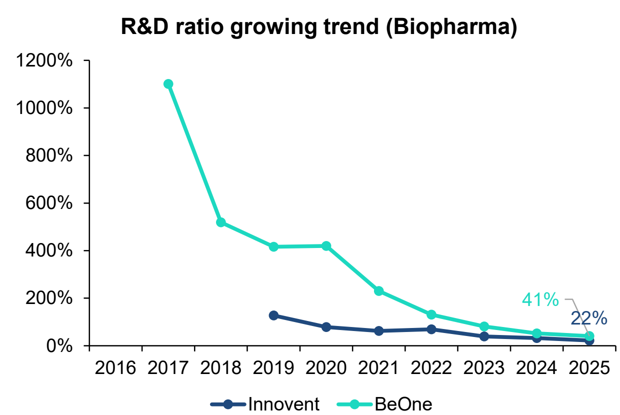

Source: Company disclosures, Bernstein analysis

EXHIBIT 8: **R&D ratio of China pharma companies have steadily increased over the years**



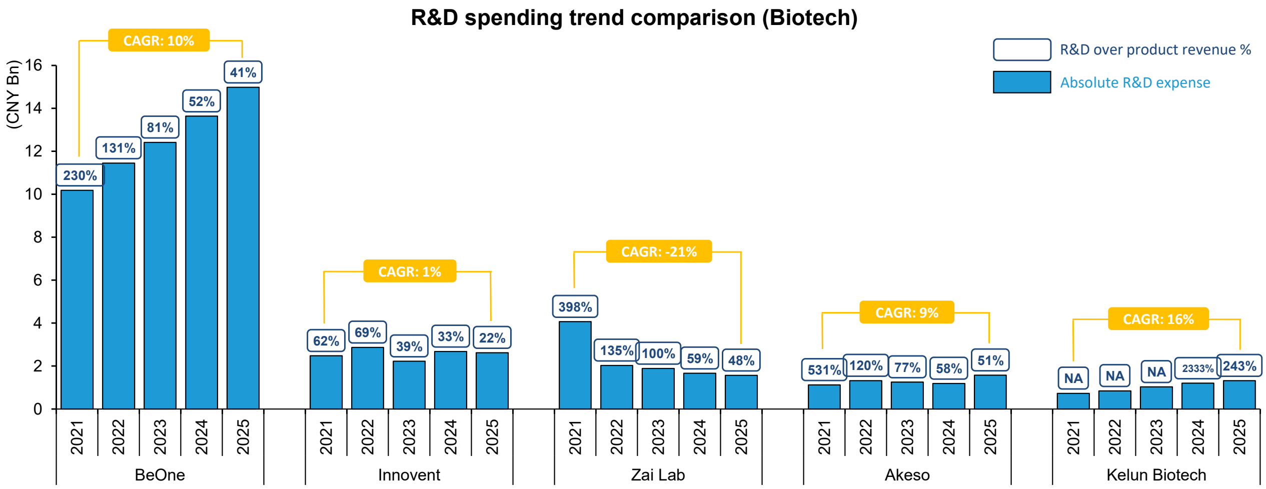

Source: Company disclosure, Bernstein analysis

EXHIBIT 9: **China biopharma companies heavily invested in R&D during early years**



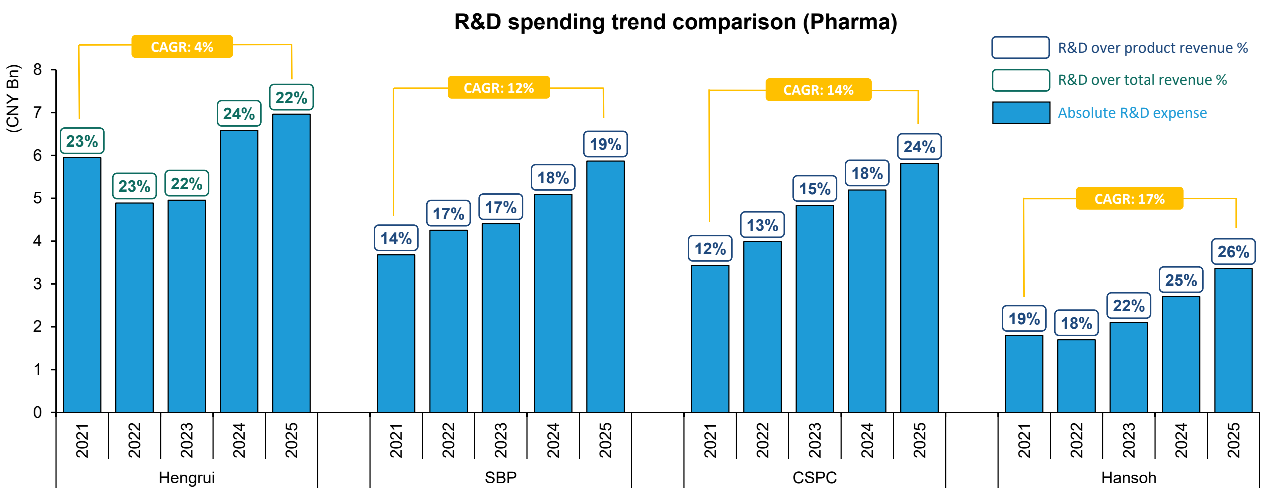

Source: Company disclosure, Bernstein analysis

EXHIBIT 10: **Commercial scaling has successfully brought R&D ratio down to more sustainable levels for biotech, with absolute R&D spending increasing across most biotech except for Zai Lab**




Source: Company disclosures, Bernstein analysis

EXHIBIT 11: **Traditional pharma are generally increasing their R&D intensity in terms of both margin and absolute spending to support drug development and clinical trials**



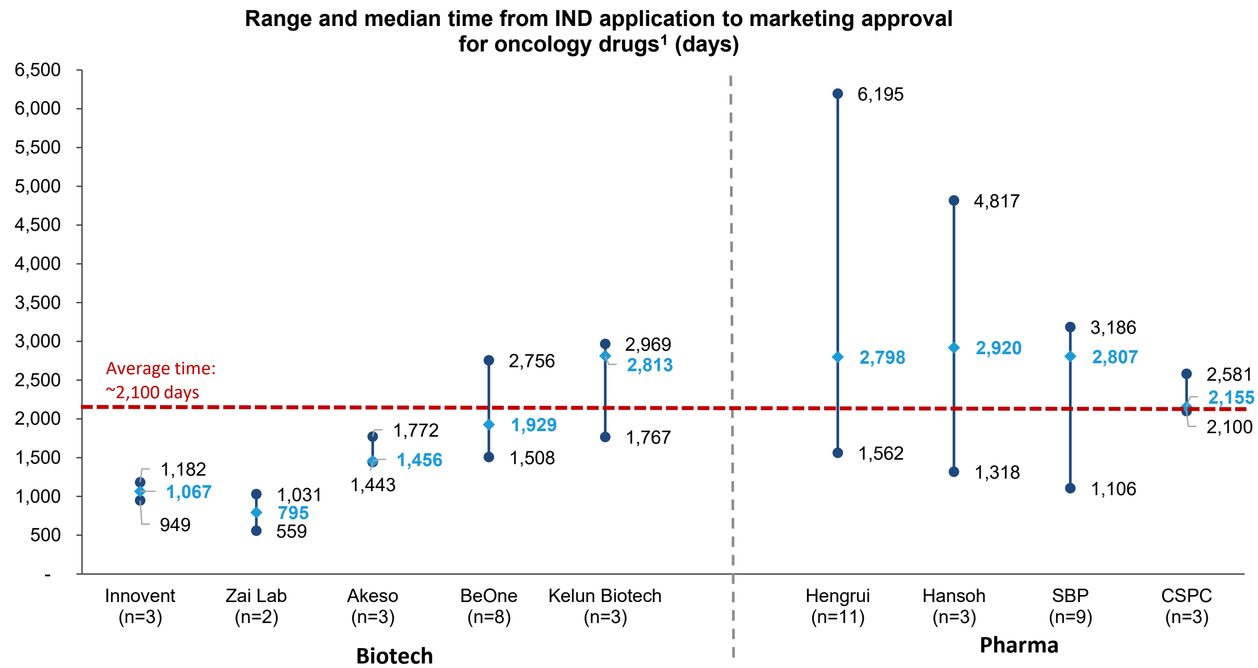

Source: Company disclosures, Bernstein analysis

## EFFICIENCY METRICS: SPEED AND CONVERSION

### TIME-TO-MARKET: FROM IND APPLICATION TO NDA APPROVAL

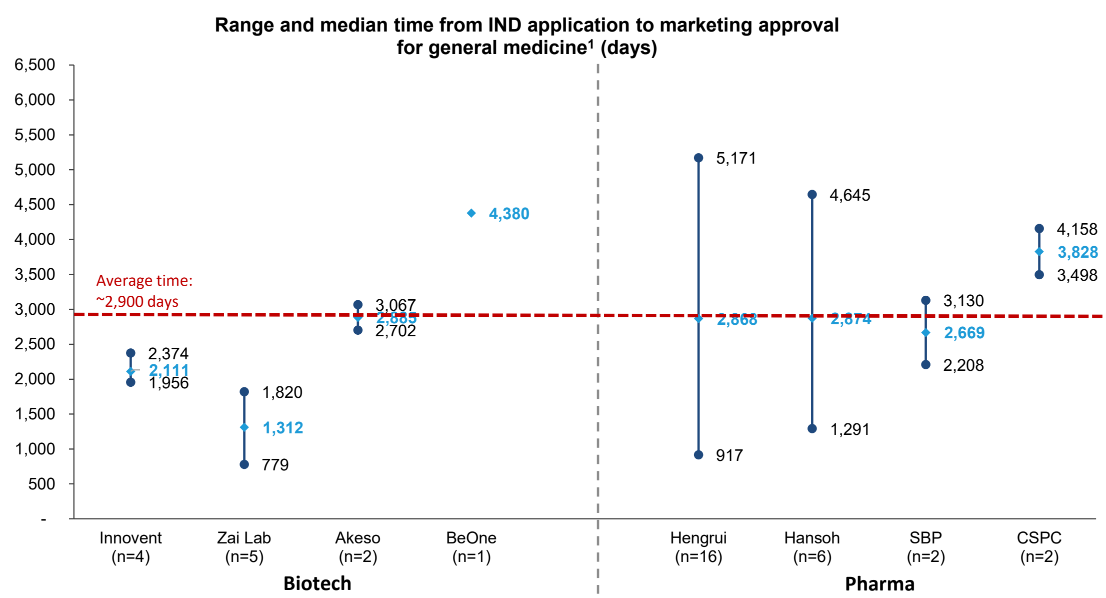
We use **time-to-market as a proxy to evaluate drug development efficiency for covered companies**, calculated as the number of days from the first IND application (CDE acceptance date) to first market authorization (CDE results release date). By TA, **oncology assets generally feature faster time-to-market vs. general medicine assets**, driven by higher unmet needs and greater regulatory focus on oncology innovations. By company type, **biotechs are structurally more development-efficient than pharma companies,** owing to agile clinical execution and more specialized focus on targeted innovative assets. Among covered companies, **Innovent consistently registers the shortest time-to-market** across both oncology and general medicine, second only to Zai Lab. Zai Lab's faster timelines stem from its in-licensing-centric business model, leveraging globally approved or late-stage imported assets. Akeso also demonstrates strong development efficiency especially within oncology (Exhibit 12-Exhibit 14).

EXHIBIT 12: **Biotech exhibits structurally shorter median innovative drug development cycles compared to traditional pharma in oncology space, with Innovent being the most rapid player**




Note: 1. Innovative drugs only
Source: Pharmcube, Bernstein analysis

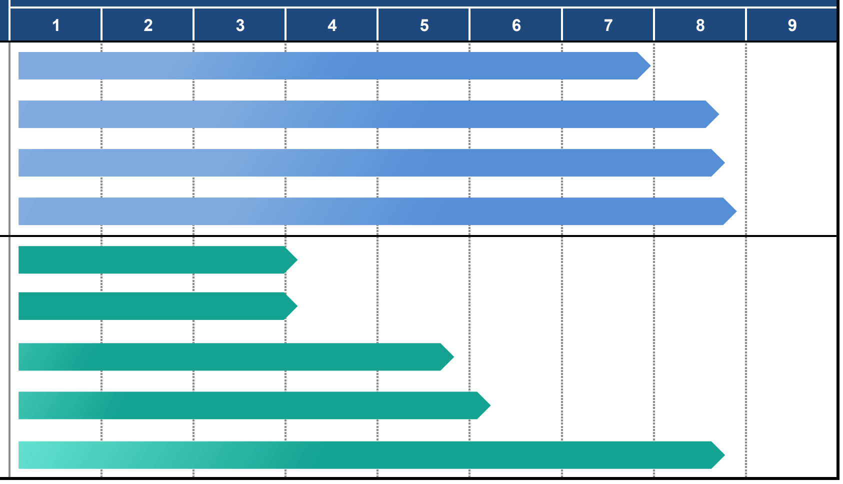
EXHIBIT 13: **Similar trend has been observed in non-oncology drugs**




Note: 1. Innovative drugs only
Source: Pharmcube, Bernstein analysis

EXHIBIT 14: **Median no. years between IND application to marketing approval; leading biotechs appear to be more efficient than pharma**

| Company type | Company | # of NMPA approved innovatives | Median time from first NMPA IND application to market approval (Year) |
|---|---|---|---|
| Pharma/ Biopharma | CSPC | 11 | |
| | Hengrui | 29 | |
| | SBP | 13 | |
| | Hansoh | 10 | |
| Biotech | Innovent | 11 | |
| | Zai Lab | 9 | |
| | Akeso | 7 | |
| | BeOne | 14 | |
| | Kelun Biotech | 3 | |

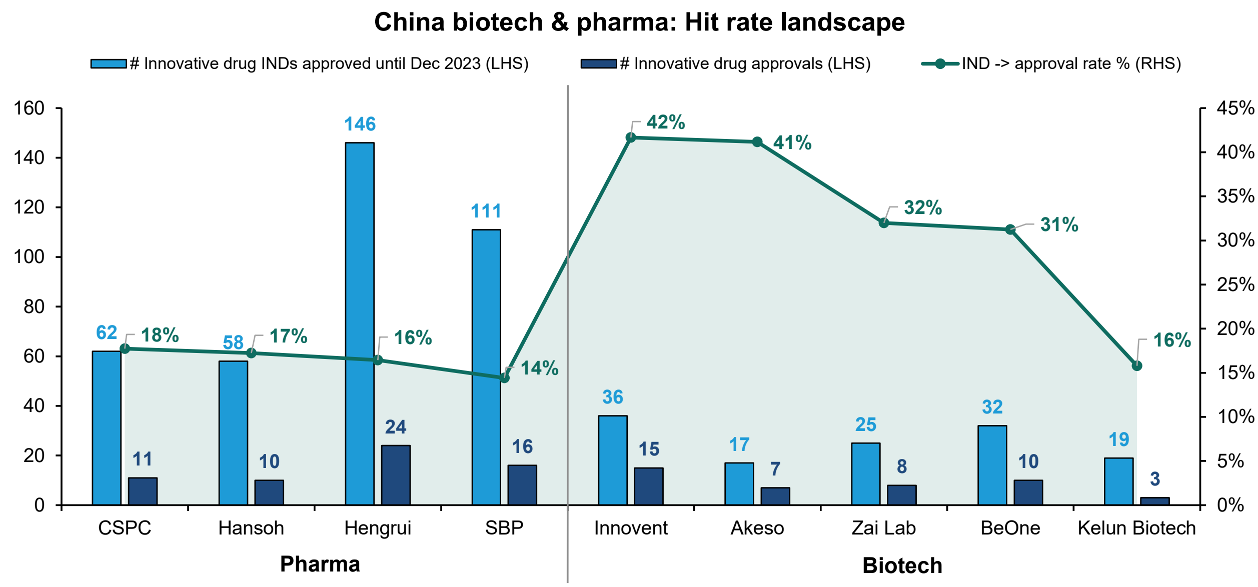



Source: Pharmcube, Bernstein analysis

## HIT RATE: % OF IND PRODUCTS GETTING APPROVED FOR MARKETING

Beyond evaluating clinical development speed, we further assess IND-to-approval hit rate as another core R&D efficiency metric, quantified as total innovative drug approved for marketing divided by cumulative innovative drug IND submissions until 2023 year-end. Results shows that biotechs achieve nearly twice the IND-to-approval hit rate compared to pharma players. At the company level, Innovent and Akeso deliver the strongest hit rates, while Hengrui and SBP have more misses with their large denominators of INDs.

EXHIBIT 15: **Biotech achieves significantly higher IND-to-approval hit rates vs. pharma despite falling short in IND volume**




Source: Pharmcube, Bernstein analysis

## BASE METRICS AND TRENDS

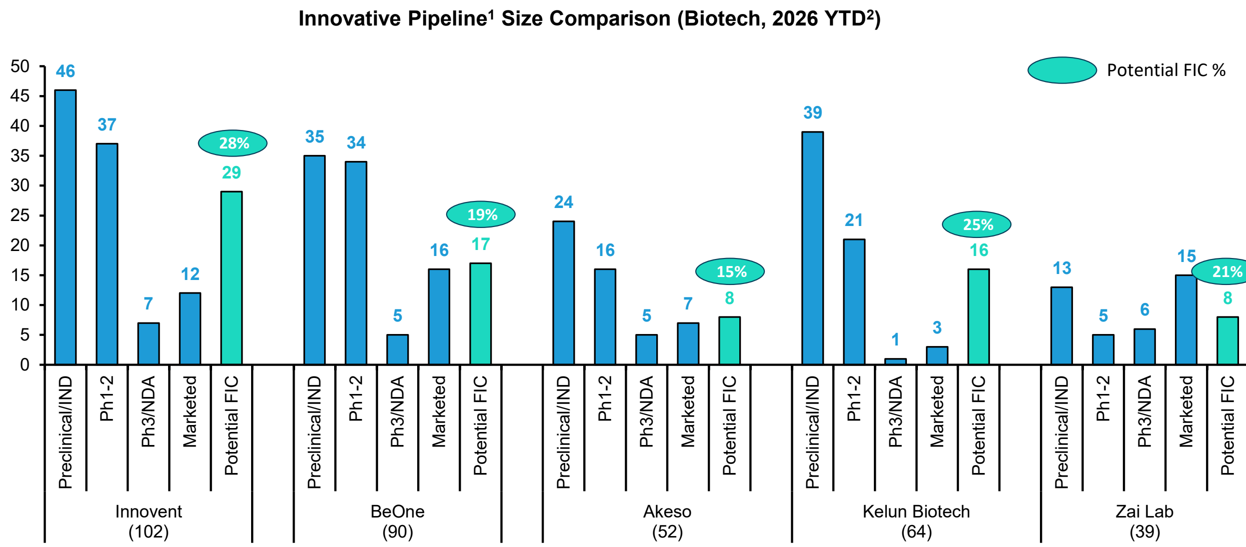
## INNOVATIVE PIPELINE SIZE

Hengrui still dominates with quantity with 360+ assets. Notably Innovent now has over 100 assets, entering the level of mature biopharma; also highest potentially FIC % at 28% vs. average c. 20%.

## TRIALS SPONSORED

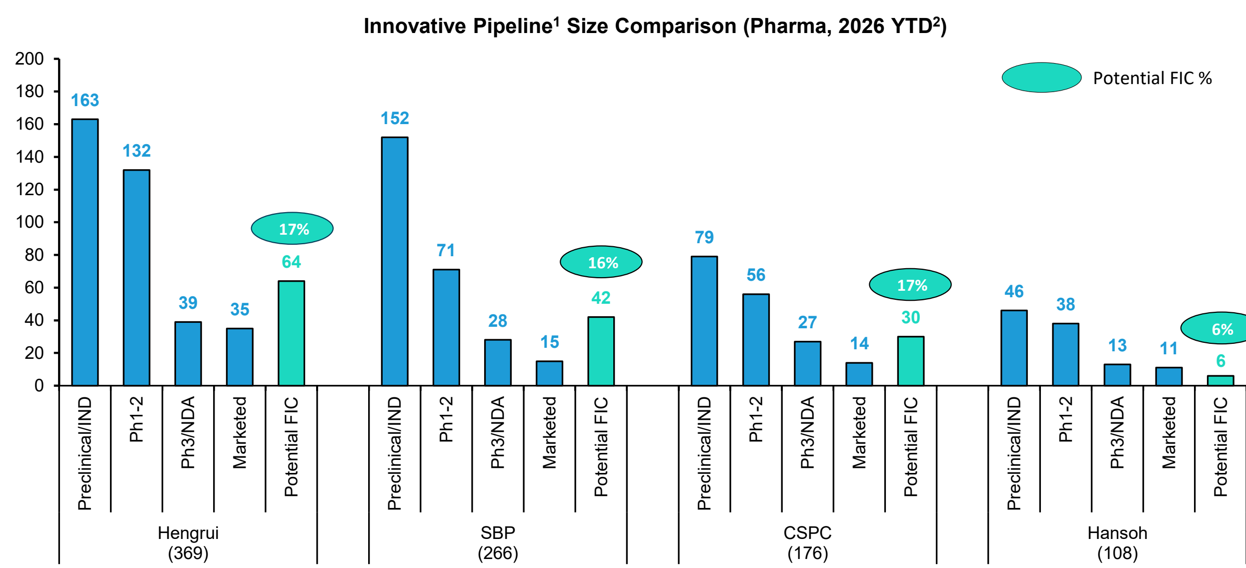
Hengrui and Innovent have sponsored the most trials so far. Interestingly, we looked at no. trials sponsored per billion RMB of R&D spending, and Akeso and Hengrui are the most efficient, averaging ~20 trials per year.

EXHIBIT 16: **Led by Innovent and BeOne in pipeline size, biotechs display high qualitative depth in their pipeline portfolio, with potential FIC assets accounting for 15-30%**



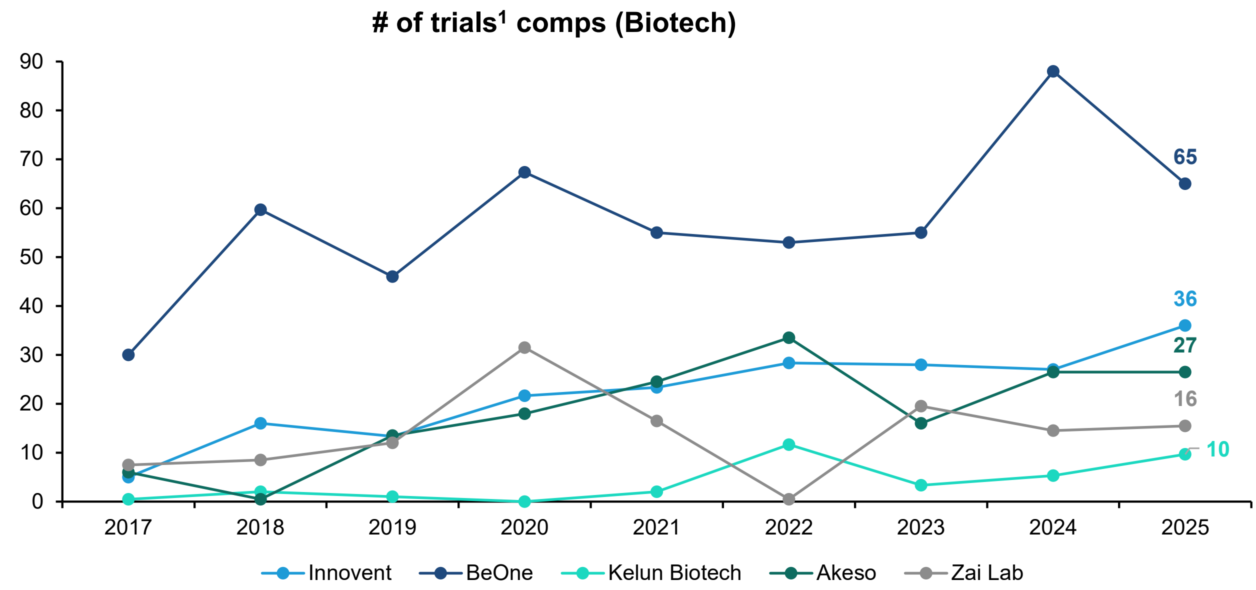

Note: 1. Include incrementally modified drugs; 2. As of May 18th 2026
Source: Pharmcube Bernstein analysis

EXHIBIT 17: **While absolute volumes are generally larger vs. biotechs, pharma's overall potential FIC exposure sits at a lower level**



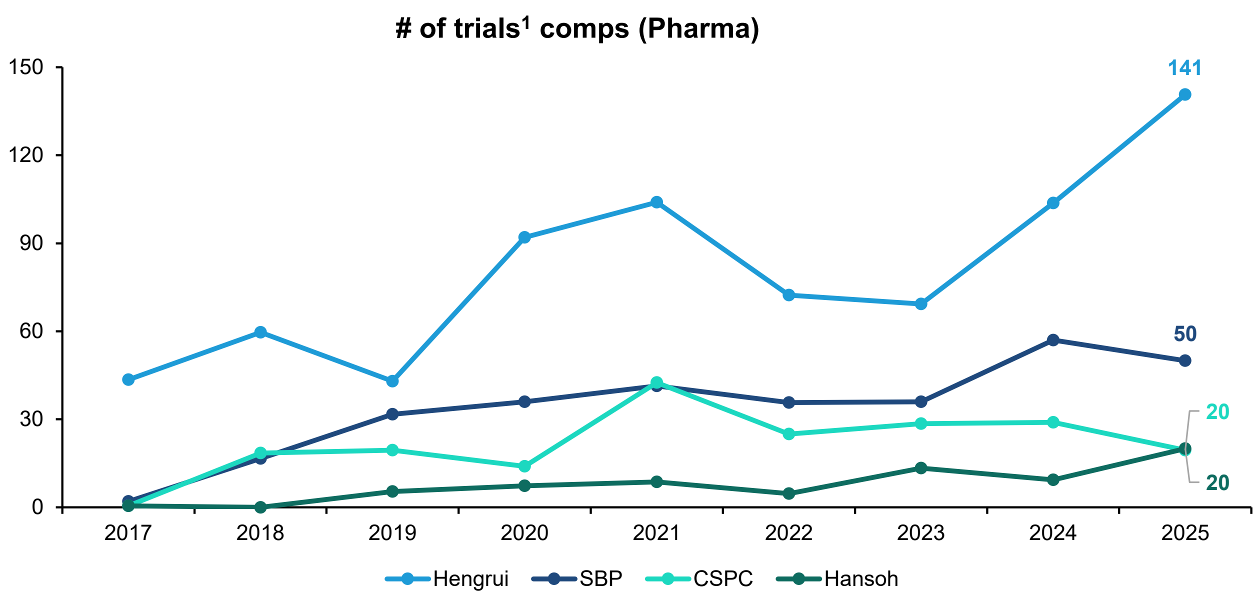

Note: 1. Include incrementally modified drugs; 2. As of May 18th 2026
Source: Pharmcube Bernstein analysis

EXHIBIT 18: **BeOne continues to command the largest clinical footprint, with Innovent and Akeso catching up steadily**



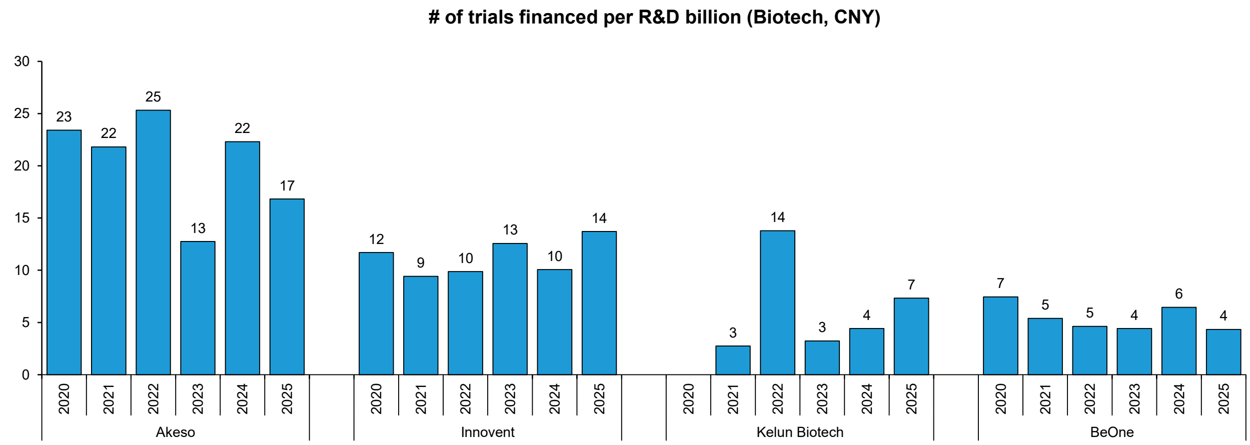

Note: 1. Composite trial count is calculated using weighted clinical trials: China p1/2 trials are counted as 0.5x, China p3 trial as 1x, and US/Australia trials as 2x
Source: Pharmcube, Bernstein analysis and estimates

EXHIBIT 19: **Hengrui has the strongest commitment in terms of clinical development, with a dominant 141 composite trials in 2025 vs. its peers' 20-50**



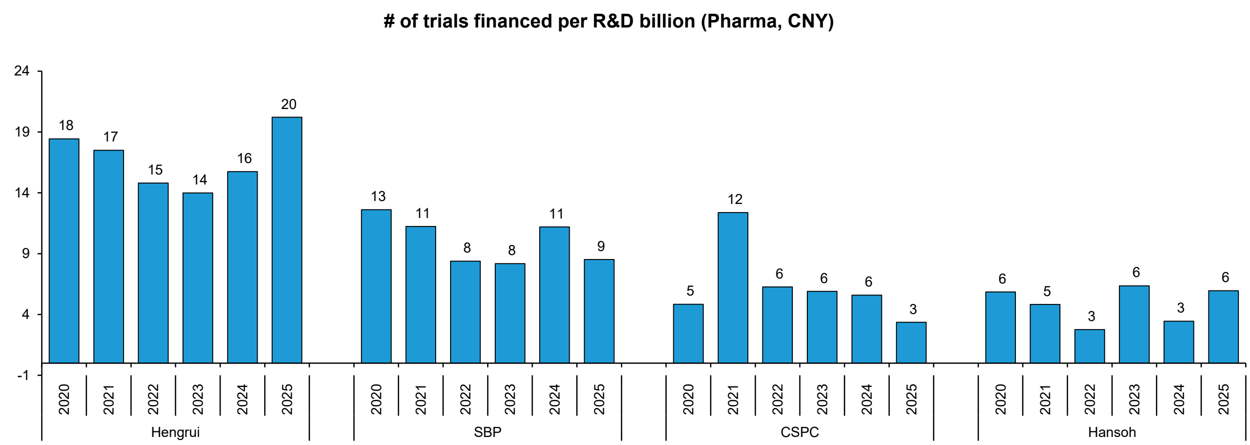

Note: 1. Composite trial count is calculated using weighted clinical trials: China p1/2 trials are counted as 0.5x, China p3 trial as 1x, and US/Australia trials as 2x
Source: Pharmcube, Bernstein analysis and estimates

EXHIBIT 20: **Akeso is the most efficient in terms of R&D-to-trial conversion among biotech peers**




Source: Company disclosures, Pharmcube, Bernstein analysis

EXHIBIT 21: **Hengrui is the most efficient in terms of R&D-to-trial conversion among pharma peers**




Source: Company disclosures, Pharmcube, Bernstein analysis

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The Bernstein brand has three categories of ratings:

- Outperform: Stock will outpace the market index by more than 15 pp
- Market-Perform: Stock will perform in line with the market index to within +/-15 pp
- Underperform: Stock will trail the performance of the market index by more than 15 pp

Coverage Suspended: Coverage of a company under the Bernstein research brand has been suspended. Ratings and price targets are suspended temporarily, are no longer current, and should therefore not be relied upon.

Not Rated: A rating assigned when the stock cannot be accurately valued, or the performance of the company accurately predicted, at the present time. The covering analyst may continue to publish research reports on the company to update investors on events and developments.

Not Covered (NC) denotes companies that are not under coverage.

Bernstein brand stock ratings are based on a 12-month time horizon.

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- Outperform (OP): Stock will outpace the relevant index by more than 10 pp
- Neutral (N): Stock will perform in line with the market index to within +/-10 pp
- Underperform (UP): Stock will trail the performance of the relevant index by more than 10 pp

Coverage Suspended: Coverage of a company under the Autonomous research brand has been suspended. Ratings and price targets are suspended temporarily, are no longer current, and should therefore not be relied upon.

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Those denoted as 'Feature' (e.g., Feature Outperform FOP, Feature Under Outperform FUP) are our core ideas.

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Autonomous brand common stock ratings are based on a 12-month time horizon.

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- Underperform (UP): The total return of the preferred instrument is expected to underperform preferred securities of other issuers operating in similar sectors or rating categories over the next six months.

Autonomous preferred stock ratings are based on a 6-month time horizon.

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